Dominant Currencies and Global Imbalances

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May 5, 2026

Roadmap

Motivation

Model Setup: 2 countries
- Rest of the World block
- Hegemon block

Global imbalances
- Four-country extension

Global imbalances are back

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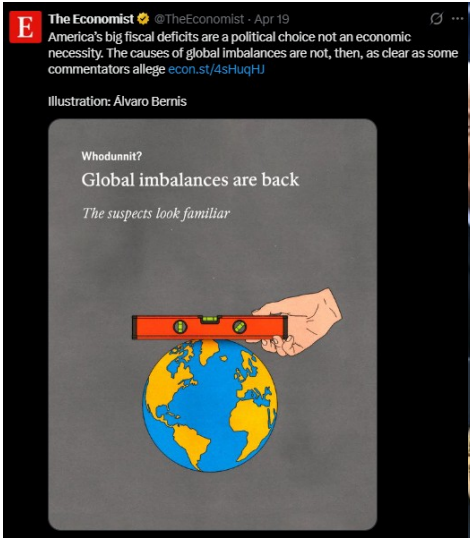
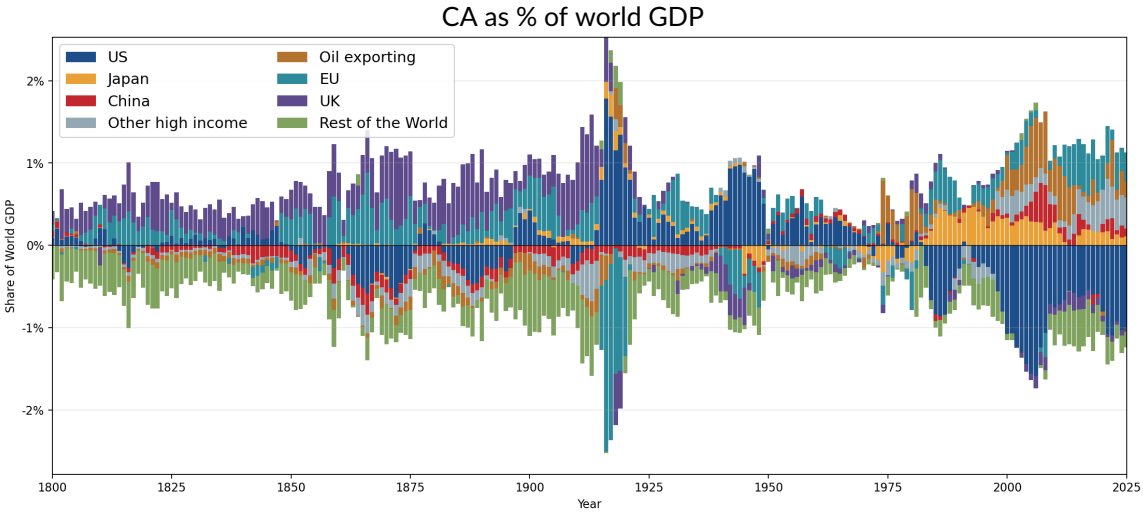


Global imbalances are back

But they never left

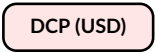


▸ FX reserves by currency ▸ Trade by invoicing currency

This Paper

- Builds a 2country model that bridges the DCP with Global Imbalances
- Revisits the Triffin Dilemma:
  - Reserve currency issuer is not obliged to run CA deficits
  - But under only 1 reserve currency issuer, the system needs liquidity
- With 4 regions and 2 reserve currency issuers:
  - DCP is not enough to replicate global imbalances
  - With currency orthogonality, reserve currency issuers always run CA deficits
  - With perfect substituability between currencies, then one hegemon can chose not to run CA deficits

The Triffin Dilemma



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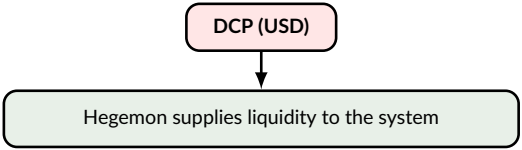

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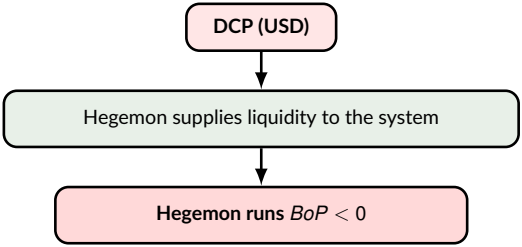

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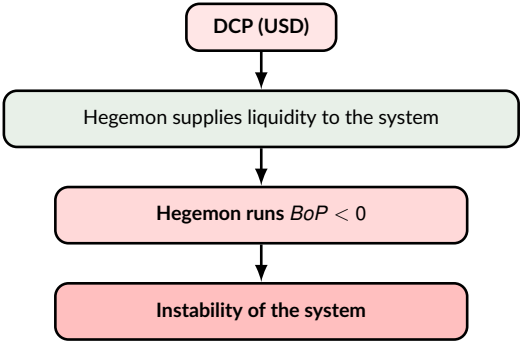

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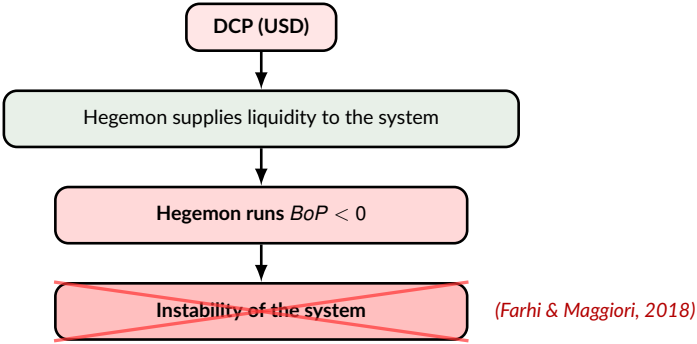
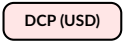


Revisiting the Triffin Dilemma



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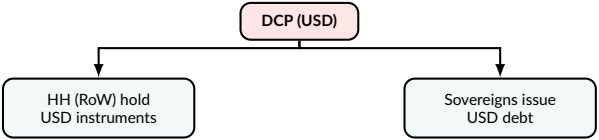

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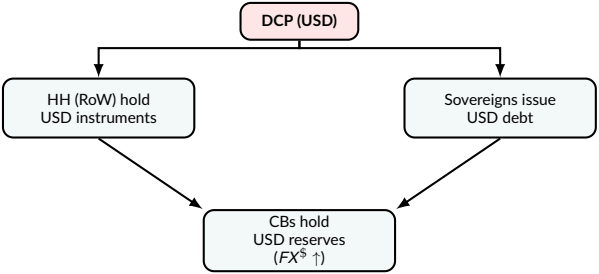

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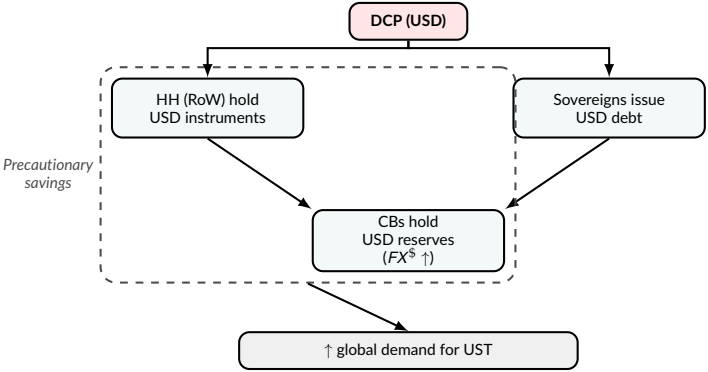

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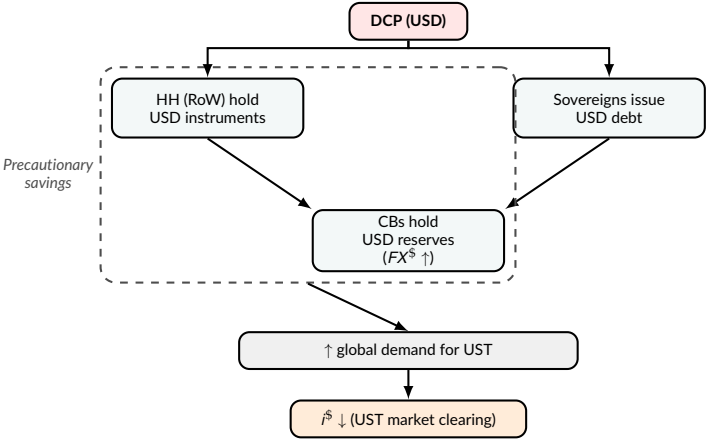

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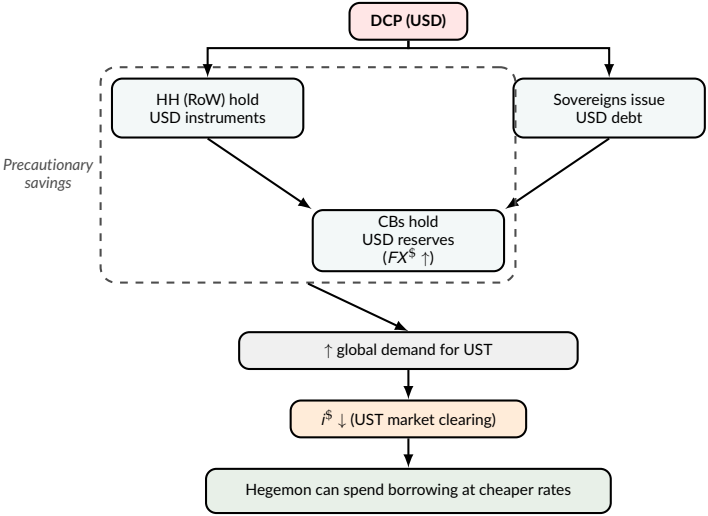

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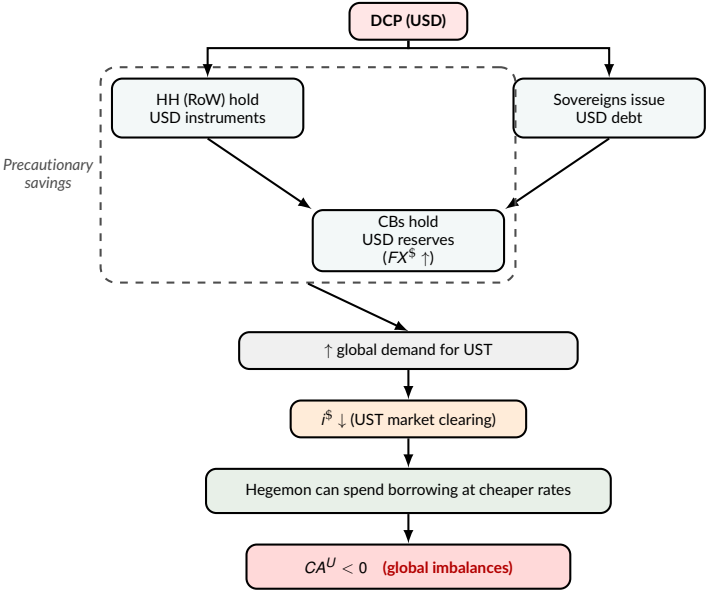

Why dominant currencies?

Why dominant currencies?

The Guardian

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Currencies

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China calls for end to dollar's reign as global reserve currency

Governor of Chinese central bank proposes new monetary system controlled by IMF

Tania Branigan *in Beijing*

Tue 24 Mar 2009 07.06 EDT

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China has called for the replacement of the dollar as the main global reserve currency, in a display of its growing assertiveness ahead of the G20 summit in London.

The bold proposal from the governor of the Chinese central bank - who calls for a new system controlled by the International Monetary Fund - reflects the willingness of emerging powers to press their case as the economic crisis overturns existing relationships and highlights the failings of the current financial system.

Zhou Xiaochuan was the PBOC governor

Why dominant currencies?

PBOC Chief Defends China's Handling of Trade Surplus

Pan Gongsheng on March 22. *Photographer: Qilai Shen/Bloomberg*

By Bloomberg News

March 22, 2026 at 2:10 AM EDT
Corrected March 23, 2026 at 10:27 PM EDT

*"When analyzing **global economic imbalances**, we should consider not only economic factors but also non-economic factors. [...] When making the analysis, we should not only analyze the international economic and trade system, but also the international monetary system. [...] major deficit countries remain the same over the years, which is related to the inherent flaws of the international monetary system.*

*Under the **international monetary system dominated by a single sovereign currency**, the country issuing the major reserve currency can sustain longstanding deficit spending with relatively low financing costs, and export its currency through sizable current account deficits."*

(emphasis and omissions added)

Related literature

- **DCP/International Monetary & Price Systems:** (Triffin, 1960; Obstfeld, 2013; Farhi & Maggiori, 2018; Eichengreen, 2019; Gopinath & Stein, 2021; Gopinath & Itskhoki, 2022; Mukhin, 2022).
- **Global Imbalances:** (Caballero, Farhi, & Gourinchas, 2008; Mendoza, Quadrini, & Ríos-Rull, 2009; Corte, Riddiough, & Sarno, 2016; Auclert, Malmberg, Martenet, & Rognlie, 2021; Caballero, Farhi, & Gourinchas, 2021; Sposi, 2022; Rey, Weder di Mauro, & Zettelmeyer, 2026; Cesa-Bianchi, Ferrero, Fornaro, & Wolf, 2026; Gourinchas, Kindberg-Hanlon, Patnam, Rotunno, & Ruta, 2026; Ma & Wei, 2026).

My contribution: Linking the role of international currencies and hegemon behaviour to global imbalances.

Roadmap

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Model Setup: 2 countries

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Global imbalances

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Asymmetric Model with a Hegemon and RoW

- Two countries i: the Hegemon (U) and RoW (R). U is small.
- Endowment economy, Y_i.
- An exogenous fraction $\alpha < 1$ of the goods purchased by RoW are priced in USD (DCP).
- Three assets:
  1. B^R a nominal bond issued by RoW in its currency
  2. $B^{R,\$}$ a nominal bond issued by RoW in U's currency
  3. $B^{U,\$}$ a nominal bond issued by the Hegemon in its currency
- Good prices are sticky in their currency of invoicing between t0 and t1.

Convention: E_t = LCU per USD; $\Delta s_{t+1} \equiv \ln E_{t+1} - \ln E_t > 0$ is LCU depreciation (USD appreciation).

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RoW household problem: the private dollar hedge target

$$\max_{\{C^R_{R,t},\, C^R_{U,t},\, B^U_{R,t},\, B^R_{R,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{(C^R_t)^{1-\gamma}}{1-\gamma} \tag{1}$$

The consumption index C_t is a Cobb–Douglas aggregator:

$$C^R_t = \left(C^R_{R,t}\right)^{1-\alpha} \left(C^R_{U,t}\right)^{\alpha}$$

With:

- $\alpha \in (0,1)$ the expenditure share on USD-priced goods.

Price index (LCU): $P_{C,t} = P^{1-\alpha}_{R,t} \left(E_t P^{\$}_{F,t}\right)^{\alpha}$, i.e., the LCU price of imports is $E_t P^{\$}_{F,t}$.

Budget constraint (LCU units): (For now, HH RoW cannot save in $B^{R,\$}$)

$$P^R_{C,t} C^R_t + E_t \cdot B^{U,USD}_{R,t} + B^R_{R,t} = P_{R,t} Y^R_t + E_t (1 + i^{USD}_{t-1}) B^{U,USD}_{R,t-1} + (1 + i^R_{t-1}) B^R_{R,t-1} - T^R_t,$$

- T^R_t : net taxes paid (taxes minus transfers)

RoW HH motivation to hold USD bonds

- A RoW depreciation, $\Delta s_{t+1} > 0$, raises the local-currency price of imports: $\Delta p^{R}_{C,t+1} \approx \alpha\, \Delta s_{t+1}$.

- If real absorption adjusts sluggishly within the period, this creates consumption risk: $\Delta c^{R,0}_{t+1} \approx -\alpha\, \Delta s_{t+1}$.

- A dollar bond pays off in local-currency terms precisely in this state: $\Delta \widetilde{R}^{\$}_{t+1} \approx \Delta s_{t+1}$.

$\Rightarrow$ Dollar assets insure RoW households against DCP-induced purchasing-power risk.

$\Rightarrow$ A higher α raises the **target demand for dollar safe assets**.

- This is the complementarity between the currency of invoicing and the currency of assets. (Gopinath & Stein, 2021)

Optimal dollar exposure

Define dollar exposure relative to consumption expenditure: $\omega_t \equiv \frac{\mathcal{E}_t B_{R,t}^{U,\$}}{P_{C,t}^R C_t^R}$.

Optimal dollar exposure

Define dollar exposure relative to consumption expenditure: $\omega_t \equiv \frac{\mathcal{E}_t B^{U,\$}_{R,t}}{P^R_{C,t} C^R_t}$.

A second-order approximation gives:

$$\boxed{\omega^d_t = \underbrace{\frac{\mu_t}{\gamma \sigma^2_{\Delta r,t}}}_{\text{return/speculative term}} + \underbrace{\frac{\alpha_t \sigma^2_{e,t} + (1-\alpha_t)\rho_t \sigma_{\pi,t}\sigma_{e,t}}{\sigma^2_{\Delta r,t}}}_{\bar{\omega}_t\text{, hedge target}}}$$

with $\mu_t \equiv i^{\$}_t - i^R_t + \mathbb{E}_t[\Delta s_{t+1}]$, $\quad \sigma^2_{\Delta r,t} \equiv \mathrm{Var}_t(\widetilde{\Delta r}_{t+1})$, $\quad \widetilde{\Delta r}_{t+1} \approx \Delta s_{t+1}$.

DCP implies $-\widehat{c}^{R,0}_{t+1} \approx \alpha_t \Delta s_{t+1} + (1-\alpha_t)\pi^H_{t+1}$, so $\mathrm{Cov}_t\left(-\widehat{c}^{R,0}_{t+1}, \widetilde{\Delta r}_{t+1}\right) \approx \alpha_t \sigma^2_{e,t} + (1-\alpha_t)\rho_t \sigma_{\pi,t}\sigma_{e,t}$.

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Under sticky domestic prices, $\rho = 0$, and if $\sigma^2_{\Delta r,t} \approx \sigma^2_{e,t}$:

$$\boxed{\bar{\omega}_t \approx \alpha_t.}$$

Interpretation

- $\bar{\omega}_t$ is the full-insurance dollar exposure.
- If $\mu_t = 0$, then $\omega^d_t = \bar{\omega}_t \approx \alpha_t$.
- If $i^\$_t + \mathbb{E}_t[\Delta s_{t+1}] < i^R_t$, then $\mu_t < 0$, so desired dollar holdings are below the full-hedge target.
- Hence α_t raises the hedge target

▸ RUIP

RoW public sector: financing need and original sin

Government budget constraint.

$$G_t^R + (1 + i_{t-1}^R)B_{t-1}^{G,R} + \mathcal{E}_t(1 + i_{t-1}^{R,\$})B_{t-1}^{R,\$} = T_t^R + B_t^{G,R} + \mathcal{E}_t B_t^{R,\$}.$$

Sovereign financing need.

$$\mathcal{F}_t^R \equiv B_t^{G,R} + \mathcal{E}_t B_t^{R,\$} = G_t^R - T_t^R + (1 + i_{t-1}^R)B_{t-1}^{G,R} + \mathcal{E}_t(1 + i_{t-1}^{R,\$})B_{t-1}^{R,\$}.$$

RoW public sector: financing need and original sin

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Currency composition of public financing. The RoW government chooses how much of $\mathcal{F}_t^R$ to finance in dollars:

$$\min_{B_t^{R,\$} \in [0, \mathcal{F}_t^R / \mathcal{E}_t]} \left\{ i_t^R B_t^{G,R} + i_t^{R,\$} \mathcal{E}_t B_t^{R,\$} + \frac{\phi_h}{2 Y_t^R} (B_t^{G,R})^2 + \frac{\phi_\$}{2 Y_t^R} (\mathcal{E}_t B_t^{R,\$})^2 \right\},$$

subject to

$$B_t^{G,R} = \mathcal{F}_t^R - \mathcal{E}_t B_t^{R,\$}.$$

Risky dollar borrowing rate. Dollar debt carries a crisis/default spread over USTs:

$$1 + i_t^{R,\$} = (1 + i_t^\$) \frac{(1 - \pi_t) + \pi_t \kappa_M}{(1 - \pi_t) + \pi_t \kappa_M (1 - L)}.$$

with $\pi_t = \bar{\pi} \exp\left(-\tilde{\xi} \frac{FX_t^\$}{N_t} \right)$, $\kappa_M =$ crisis-state pricing weight in the risky dollar borrowing rate

RoW public sector: financing need and original sin

Sovereign financing need.

$$\mathcal{F}_t^R \equiv B_t^{G,R} + \mathcal{E}_t B_t^{R,\$} = G_t^R - T_t^R + (1 + i_{t-1}^R) B_{t-1}^{G,R} + \mathcal{E}_t (1 + i_{t-1}^{R,\$}) B_{t-1}^{R,\$}.$$

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Closed-form dollar-financing target

$$\boxed{\mathcal{E}_t B_t^{R,\$,*} = \left[\frac{Y_t^R (i_t^R - i_t^{R,\$}) + \phi_h \mathcal{F}_t^R}{\phi_h + \phi_\$} \right]_0^{\mathcal{F}_t^R}}$$

Interpretation: RoW finances its public need with local-currency debt and dollar debt. Dollar debt is attractive when dollar funding is cheap, but it carries original-sin risk. The optimal dollar share equates marginal local-funding costs with marginal dollar-funding and crisis costs.

RoW public sector: FX reserves and UST demand

Dollar needs

After the sovereign chooses the currency composition of financing, dollar debt enters dollar needs:

$$N_t = B_t^{R,\$} + \chi_M \alpha C^R$$

Reserve motive

Given dollar needs, the RoW public sector chooses dollar reserves to insure against crisis risk:

$$\min_{FX_t^\$ \geq 0} \left\{ (i_t^R - i_t^\$) FX_t^\$ + \Lambda \pi_t \right\}.$$

with $\pi_t = \bar{\pi} \exp\left(-\zeta \frac{FX_t^\$}{N_t} \right)$

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$$\min_{FX_t^{\$} \geq 0} \left\{ (i_t^R - i_t^{\$}) FX_t^{\$} + \Lambda \pi_t \right\}.$$

From the reserve FOC

$$i_t^R - i_t^{\$} = \Lambda \frac{\xi}{N_t} \pi_t.$$

Equivalently:

$$\frac{FX_t^{\$}}{N_t} = \frac{1}{\xi} \log \left(\frac{\Lambda \xi \bar{\pi}}{(i_t^R - i_t^{\$}) N_t} \right).$$

with $\xi =$ effectiveness of reserves, $\Lambda =$ cost/severity of crisis

Interpretation

- Higher α_t raises the dollar import bill $M_t^{\$}$.
- Higher RoW public financing needs raise $\mathcal{F}_t^R$.
- The funding-composition rule determines how much of $\mathcal{F}_t^R$ is financed in dollars.
- Higher $B_t^{R,\$}$ and higher $M_t^{\$}$ raise dollar needs N_t.
- Higher N_t raises the scale of precautionary reserve demand.
- A larger spread $i_t^R - i_t^{\$}$ raises the opportunity cost of reserves and lowers reserve coverage.

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US fiscal policy and UST market clearing

Government budget constraint

$$G_t^U + (1 + i_{t-1}^{USD}) B_{t-1}^{U,gov} = T_t^U + B_t^{U,gov}$$

Spending financed: $G_t^U = T_t^U + \Delta B_t^{net}$.

US fiscal policy and UST market clearing

Government budget constraint

$$G_t^U + (1 + i_{t-1}^{USD}) B_{t-1}^{U,gov} = T_t^U + B_t^{U,gov}$$

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Safe-rate determination Let

$$Z_t(i, \alpha, B^{gov}) = \underbrace{B_t^{U,gov}}_{\text{Treasury supply}} - \underbrace{B_t^{U,\$}(US; i)}_{\text{US HH demand}} - \underbrace{B_t^{U,\$}(RoW; i, \alpha)}_{\text{RoW private hedge demand}} - \underbrace{FX_t^{\$}(i, \alpha)}_{\text{official UST reserves}} .$$

Market clearing is:

$$Z_t(i_t^{USD}, \alpha_t, B_t^{U,gov}) = 0.$$

US fiscal policy and UST market clearing

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Market clearing is:

$$Z_t(i_t^{USD}, \alpha_t, B_t^{U,gov}) = 0.$$

Holding Treasury supply fixed:

$$\boxed{\left.\frac{\partial i_t^{USD}}{\partial \alpha_t}\right|_{B^{U,gov}} = -\frac{\partial_\alpha B_t^{U,\$}(RoW) + \partial_\alpha FX_t^{\$}}{\partial_i B_t^{U,\$}(US) + \partial_i B_t^{U,\$}(RoW) + \partial_i FX_t^{\$}} < 0}$$

Balance of Payments: Hegemon

Trade balance and current account

$$TB_t^U = Y_t^U - C_t^U - G_t^U, \qquad CA_t^U = TB_t^U + NI_t^U$$

Net foreign assets

$$\boxed{NFA_t^U \equiv B_t^{R,\$} - B_{R,t}^{U,\$}}$$

Balance of payments identity

$$\boxed{CA_t^U = \Delta NFA_t^U}$$

RoW is the mirror: $NFA_t^R = -NFA_t^U$, $CA_t^R = -CA_t^U$, $TB_t^R = -TB_t^U$.

With one-period par bonds and no valuation effects, $VAL_t^U \simeq VAL_t^R \simeq 0$, *so* $CA_t^i = \Delta NFA_t^i$ *for* $i \in \{U, R\}$.

Roadmap

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Simulations: per-period fixed point

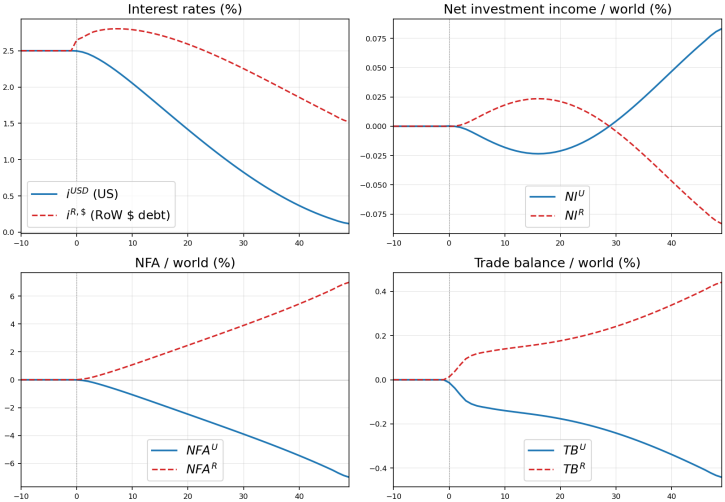
Before t_0 both countries in autarky: $\alpha = 0$, no DCP demand, $i^{USD} = i^R = g$.
Opening in $t = 0$**:** Dollar-invoicing share $\alpha^{\$}$ rises.

Each period t **jointly solves:**

1. **RoW HH:** consumption from Euler; private UST hedge B^{priv}
2. **RoW sovereign:** currency mix of new debt $\{B^{R,\$}, B^{G,R}\}$
3. **RoW CB:** chooses reserves $FX^{\$}$
4. **US fiscal:** Treasury issuance B^{gov}
5. **US HH:** consumption C^U pinned by the BoP/NFA-consistency identity
6. **UST market clears:** $B^{gov} = B^{dom} + B^{priv} + F^T$, which determines i^{USD}

Steps 1–6 are solved simultaneously: the solver iterates on (i^{USD}, C^R) *until all conditions hold.*

Baseline transition



▸ Calibration: assigned ▸ Calibration: blocks ▸ Structural

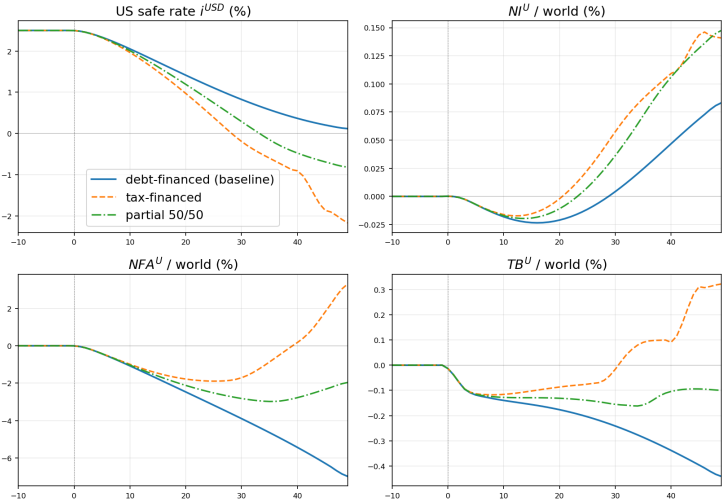
Three counterfactual exercises

- **Spending exercises:** fix the dollar-invoicing path and vary how the hegemon *finances* additional spending:
  - Debt financed
  - Tax financed
  - 50/50

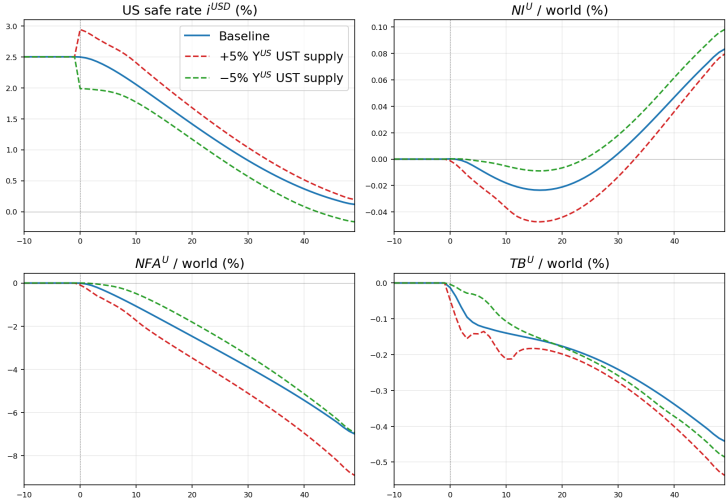
- **Exogenous UST supply:** fix the fiscal block and shift the *level* of outstanding UST

- **Stress scenarios:**
  - no-DCP regime (foreign hedge demand turned off)
  - no US spending

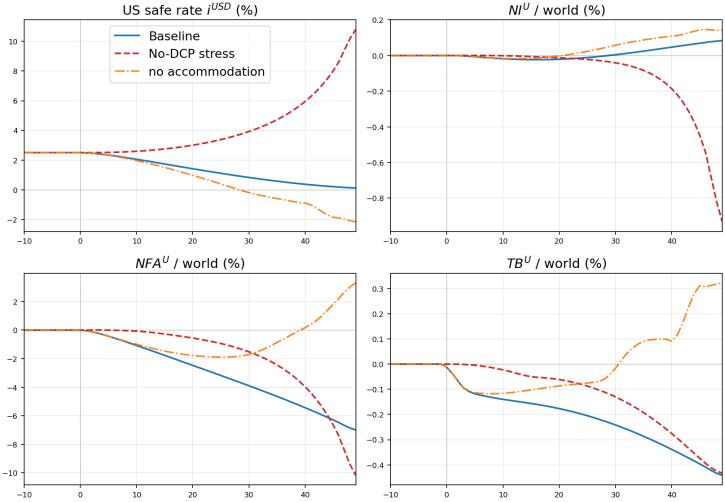
Spending exercises



Exogenous UST supply



Stress scenarios: no-DCP and no fiscal accommodation



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Motivation

Model Setup: 2 countries
- Rest of the World block
- Hegemon block

Global imbalances
- Four-country extension

Four-country benchmark: EU as second safe-asset issuer

4 regions: US (issues dollar safe assets), EU (issues euro safe assets), China and RoW.

- $i^{\$}$ and i^{E} are pinned by separate market-clearing conditions

Peripheral blocs. For $j \in \{C, R\}$, public financing is split across local, dollar, and euro debt:

$$B_{j,t}^{LCU} + B_{j,t}^{USD} + B_{j,t}^{EUR} = \mathcal{F}_{j,t}.$$

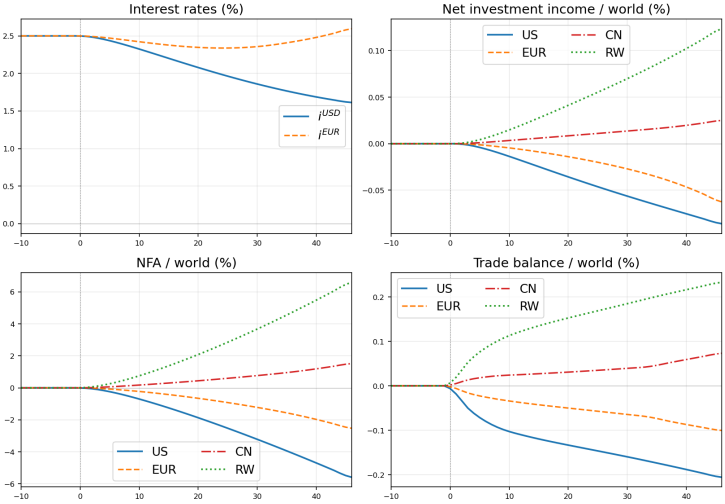
Currency-specific DCP exposure creates currency-specific safe-asset demand:

$$\alpha_{j,USD} \uparrow \Rightarrow \omega\$_{j,t}, FX_{j,t}^{\$} \uparrow, \qquad \alpha_{j,EUR} \uparrow \Rightarrow \omega_{j,t}^{E}, FX_{j,t}^{E} \uparrow .$$

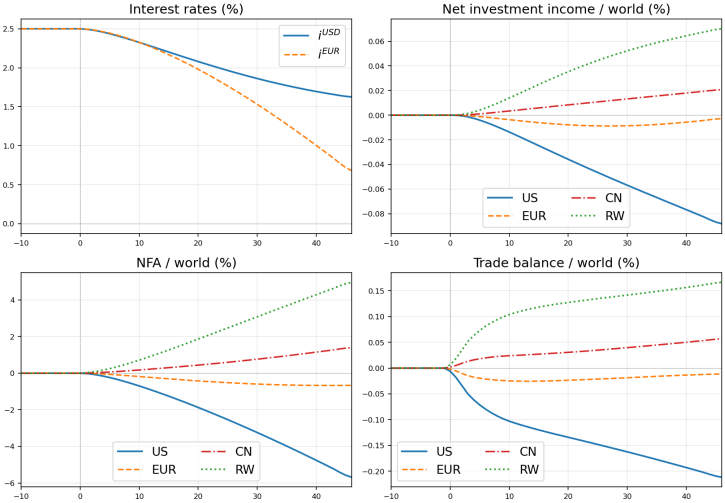
$\alpha_{j,USD} > \alpha_{j,EUR}$, $\quad FX_{jU}$ insures USD needs only, $\quad FX_{jE}$ insures EUR needs only,
The ECB holds only trade-based USD reserves.

EU fiscal block: In the benchmark, Europe behaves like the US.

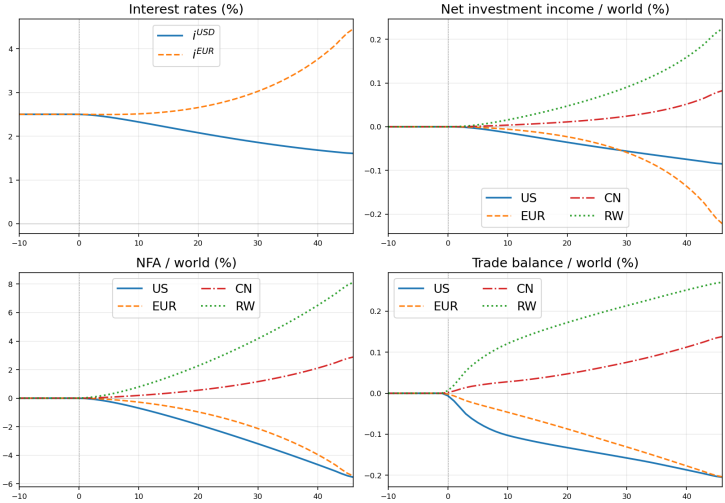
DCP is not enough to explain global imbalances



Four-country transition: Germany variant (fiscal balance)



Four-country transition: France variant (higher EU spending)



Pooled 1-for-1 regime: common hard-currency service

USTs and euro safe assets provide the same hard-currency service.

Pooled hard-currency exposure. For each peripheral bloc $j \in \{C, R\}$, total hard-currency exposure is:

$$\alpha_{jH} = \alpha_{jU} + \alpha_{jE}.$$

Aggregate hard-currency capacity. Total hard-currency safe-asset supply is:

$$S_t^H = B_t^{U,gov} + B_t^{E,gov}.$$

Total foreign hard-currency demand is:

$$D_t^H = \sum_{j \in \{C,R\}} \left(A_j^H + FX_j^H \right) + FX_{ECB}^H.$$

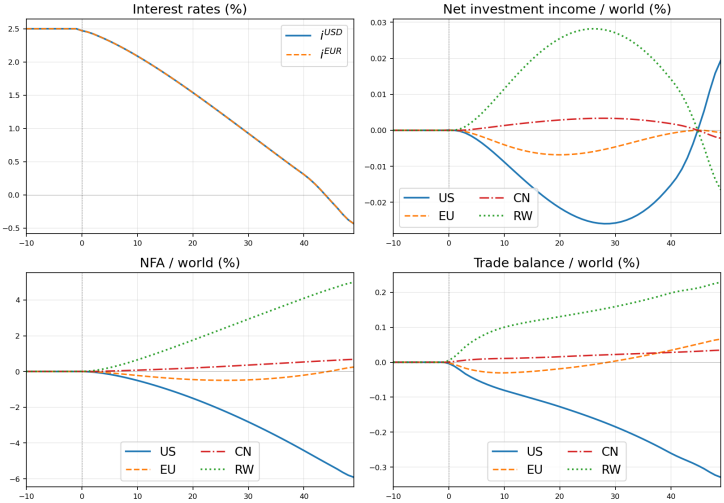
USD and EUR safe assets converge to one common hard-currency return:

$$i_t^{USD} = i_t^{EUR}$$

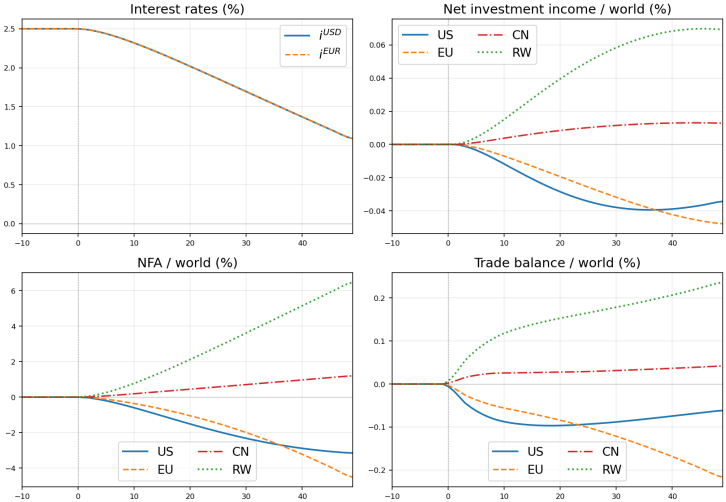
whenever both assets are supplied and traded. If one asset has zero supply, its rate is not a traded market rate:

$$B_t^{U,gov} = 0 \Rightarrow i_t^{USD} \text{ is shadow only}, \qquad B_t^{E,gov} = 0 \Rightarrow i_t^{EUR} \text{ is shadow only}.$$

EU runs fiscal surplus



US does not spend



Conclusion and next steps

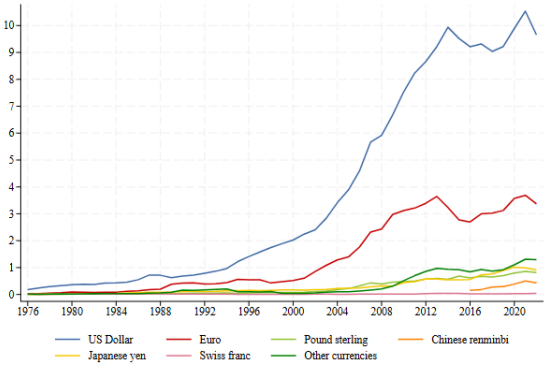
- I build a model that bridges DCP and Global Imbalances
  - Revisits the Triffin dilemma: hegemon is not forced to run CA deficits but, if no one else supplies liqudity to the system, then there is no currency to trade
  - DCP is not enough to explain current imbalances

- Next steps:
  - Understanding better the liquidity constraints and currency switching
  - Other extensions (e.g. endogenous E, separating currency of invoicing from physical trade)

Gracias!

Central Bank reserves by currency (trillions of USD)

Central Bank Reserves in trillons of USD



Source: IMF Annual Reports (1984, 1986-1988, 1990, 1999) and IMF Currency Composition of Official Foreign Exchange Reserves (COFER) (1995-2022). Deutsche marks, French francs, Dutch guilders and ECUs are included in the Euro before 1999.

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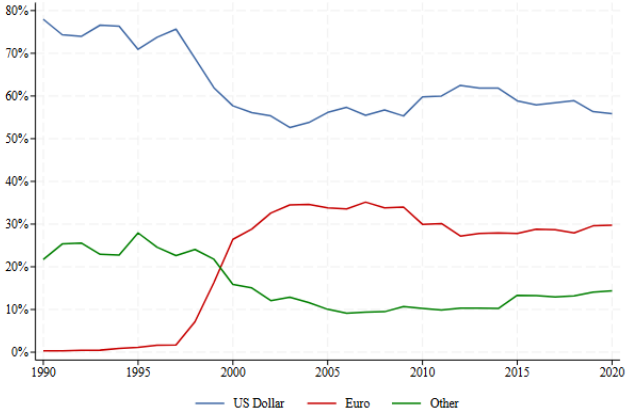
Global goods trade by invoicing currency



Author's calculations relying Boz et al. (2020). EUR includes legacy currencies. We impute regional averages for countries with missing data.

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Risk-Adjusted UIP: dollar assets as insurance

Date-t SDF

$$M^R_{t,t+1} = \beta \left(\frac{C^R_{t+1}}{C^R_t} \right)^{-\gamma}, \qquad m^R_{t+1} = \ln \beta - \gamma \Delta c^R_{t+1}.$$

m^R_{t+1} is high in bad states: low C^R_{t+1} means high marginal utility.

Cross-asset Euler equations imply risk-adjusted UIP

$$i^R_t - i^\$_t = \mathbb{E}_t[\Delta s_{t+1}] + \underbrace{\mathrm{Cov}_t \left(m^R_{t+1}, \Delta s_{t+1} \right)}_{\text{insurance value of dollar payoff}} .$$

- If the dollar appreciates in states where RoW marginal utility is high, dollar assets are valuable insurance.
- A lower dollar yield can then be sustained as an insurance premium, not as an arbitrage violation.

Under sticky domestic and foreign-currency prices and sluggish consumption :

$\mathrm{Cov}_t(\pi^H, \Delta s) = 0$, $\mathrm{Cov}_t(\pi^{F,\$}, \Delta s) = 0$, $\Rightarrow \mathrm{Cov}_t(m^{R,0}, \Delta s) \approx \gamma \alpha \sigma^2_e > 0$.

This covariance is the **background hedge motive**.

Risk-Adjusted UIP: background DCP covariance

DCP creates background purchasing-power risk before portfolio payoffs.

$$\pi^C_{t+1} = (1-\alpha)\pi^H_{t+1} + \alpha\left(\pi^{F,\$}_{t+1} + \Delta s_{t+1}\right).$$

If real consumption adjusts sluggishly to CPI movements,

$$\Delta c^{R,0}_{t+1} \approx -\pi^C_{t+1}, \qquad m^{R,0}_{t+1} \approx -\gamma \Delta c^{R,0}_{t+1}.$$

Then

$$\mathrm{Cov}_t(m^{R,0}, \Delta s) = -\gamma\,\mathrm{Cov}_t(\Delta c^{R,0}, \Delta s) \approx \gamma\,\mathrm{Cov}_t(\pi^C, \Delta s),$$

so

$$\mathrm{Cov}_t(m^{R,0}, \Delta s) \approx \gamma\Big[\alpha\ \mathrm{Var}_t(\Delta s) + (1-\alpha)\,\mathrm{Cov}_t(\pi^H, \Delta s) + \alpha\,\mathrm{Cov}_t(\pi^{F,\$}, \Delta s)\Big].$$

Benchmark case: sticky domestic and foreign-currency prices

$$\mathrm{Cov}_t(\pi^H, \Delta s) = 0, \qquad \mathrm{Cov}_t(\pi^{F,\$}, \Delta s) = 0,$$

$$\Rightarrow \mathrm{Cov}_t(m^{R,0}, \Delta s) \approx \gamma\alpha\sigma_e^2 > 0.$$

- This covariance is the **background hedge motive**.
- It shifts the target demand for dollar assets.
- The equilibrium spread is determined by Treasury-market clearing, not imposed as a separate equation.

Back

Why does a higher α raise $\mathrm{cov}_t(m, \Delta s)$? I

Starting point (annual, GS-style): With standard CRRA preferences and the purchasing-power channel,

$$\mathrm{cov}_t(m, \Delta s) \;\approx\; \gamma\, \mathrm{cov}_t(\pi^C, \Delta s),$$

and CPI inflation satisfies

$$\pi^C_{t+1} = (1-\alpha)\pi^H_{t+1} + \alpha\big(\pi^{F,\$}_{t+1} + \Delta s_{t+1}\big).$$

General expression: Taking covariance with Δs_{t+1},

$$\mathrm{cov}_t(\pi^C, \Delta s) = \alpha\, \mathrm{var}_t(\Delta s) + (1-\alpha)\mathrm{cov}_t(\pi^H, \Delta s) + \alpha\, \mathrm{cov}_t(\pi^{F,\$}, \Delta s).$$

Effect of α: Differentiating with respect to α,

$$\frac{\partial}{\partial \alpha}\mathrm{cov}_t(\pi^C, \Delta s) = \mathrm{var}_t(\Delta s) - \mathrm{cov}_t(\pi^H, \Delta s) + \mathrm{cov}_t(\pi^{F,\$}, \Delta s).$$

Cauchy–Schwarz bound. For any two random variables X, Y,

$$|\mathrm{cov}(X, Y)| \le \sqrt{\mathrm{var}(X)\mathrm{var}(Y)}.$$

Why does a higher α raise $\mathrm{cov}_t(m, \Delta s)$? II

Hence,

$$|\mathrm{cov}_t(\pi^H, \Delta s)| \leq \sigma_\pi \, \sigma_e, \qquad |\mathrm{cov}_t(\pi^{F,\$}, \Delta s)| \leq \sigma_F \, \sigma_e,$$

where $\sigma_e^2 = \mathrm{var}_t(\Delta s)$ and $\sigma_\pi^2 = \mathrm{var}_t(\pi^H)$.

Sufficient condition for monotonicity. If foreign USD inflation is weakly related to the RoW exchange rate

$$\mathrm{cov}_t(\pi^{F,\$}, \Delta s) \approx 0,$$

and if annual exchange-rate volatility dominates domestic inflation volatility,

$$\sigma_\pi \leq \sigma_e,$$

then by Cauchy–Schwarz

$$\mathrm{cov}_t(\pi^H, \Delta s) \leq \sigma_\pi \sigma_e \leq \sigma_e^2 = \mathrm{var}_t(\Delta s),$$

which implies

$$\frac{\partial}{\partial \alpha} \mathrm{cov}_t(\pi^C, \Delta s) > 0 \quad \Rightarrow \quad \frac{\partial}{\partial \alpha} \mathrm{cov}_t(m, \Delta s) > 0.$$

Economic intuition.

- Increasing α raises the weight on the *direct* exchange-rate component of CPI inflation.

Why does a higher α raise $\mathrm{cov}_t(m, \Delta s)$? III

- Unless home inflation co-moves with depreciation *more strongly than allowed by Cauchy–Schwarz*, the direct import-price channel dominates.
- Empirically, annual exchange-rate volatility typically exceeds inflation volatility, making this condition plausible.

Remark. This is a *sufficient but not necessary* condition. Monotonicity in α may fail in extreme cases, even though $\mathrm{cov}_t(m, \Delta s) > 0$ still holds.

Calibration: assigned parameters

Parameter	Value	Interpretation
β	0.97561	household discount factor
γ_{rra}	3.0	risk aversion
g	2.5%	common trend growth
Y^U	0.27	US share of world GDP
Y^R	0.73	RoW share of world GDP
$i_0^{USD} = i_0^R$	2.5%	autarky safe-rate anchor
$\alpha_{U,1}$	0.04	bilateral US–RoW physical exposure
$\alpha_{X,1}$	0.16	within-RoW dollar-invoiced exposure
$\alpha_1^{\$} = \alpha_{U,1} + \alpha_{X,1}$	0.20	total DCP exposure
σ_e	0.18	exchange-rate volatility
ρ	0	CPI/exchange-rate covariance benchmark
T_{trans}	50	transition horizon
T_{ramp}	50	DCP ramp length
H_{aux}	10	rolling auxiliary horizon

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Calibration: financial, reserve, and fiscal blocks

Parameter	Value	Role
Λ	0.70	reserve-loss weight
Λ_B	2.00	foreign-currency debt risk weight
$\bar{\pi}$	0.07	zero-reserve crisis probability
ξ	4.0	reserve-coverage sensitivity
L	0.40	loss given default
κ_M	1.50	crisis-state marginal-utility wedge
χ_M	1/4	import-bill coverage need
ϕ_A	0.25	RoW private UST demand slope
ϕ_U	0.10	US domestic UST demand slope
κ_A	2.0	private portfolio adjustment cost
κ_{FX}	3.5	reserve adjustment cost
$\kappa_{B^{R,\$}}$	4.0	RoW dollar-debt adjustment cost

RoW fiscal parameter	Value	Role
G_R/Y_R	0.14	RoW public spending
T_R/Y_R	0.11	RoW tax revenue
ϕ_h	0.50	local-currency debt convex cost
$\phi_\$$	0.50	dollar-debt convex cost
$\rho_{i^{R,\$}}$	0.97	smoothing of risky dollar spread

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From financial side closure to structural

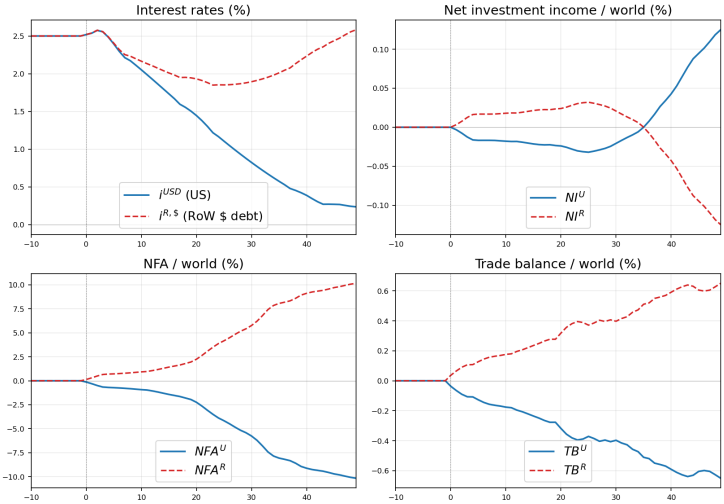
Baseline: The financial side is fully spelled out — UST market clearing, RoW funding-composition rule, dollar-spread pricing. But the household side is reduced-form: C^U is pinned by a balance-of-payments / NFA accounting identity rather than by an explicit Euler equation. The model is internally consistent on the financial side; the household side is implicit.

What the structural-HH version adds.

- Explicit U.S. household Euler equation: $(1 + i_t^{USD}) = (1/\beta)(C_{t+1}^U / C_t^U)^\gamma$.
- Explicit RoW Euler equation: $(1 + i_t^R) = (1/\beta)(C_{t+1}^R / C_t^R)^\gamma$.
- Joint UST-allocation rule: B^{priv} and F^T allocated together over scarce UST supply via weighted least-squares.
- KKT-corrected reserve FOC: handles the $i^R \leq i^\$$ corner where reserves want the cap.
- RoW dollar debt $B^{R,\$}$ and US households' UST holdings $B^{U,US}$ both endogenous, subject to no-leverage on US side.

Why it matters. The structural version is the version where Euler residuals, NFA-law residuals, and UST-clearing residuals all close *jointly*. It's the cleanest pass model.

Baseline transition: structural-HH



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Appendix: $\dfrac{\partial \omega^*}{\partial \alpha}$ and dominance of the risk–denominator term I

Object.

$$\omega^*(\alpha) = \underbrace{\frac{E_t[\Delta r_{t+1}]}{\gamma\,\sigma^2_{\Delta r}(\alpha)}}_{\text{speculative}} + \underbrace{(\gamma\kappa - 1)\,\frac{H(\alpha)}{\sigma^2_{\Delta r}(\alpha)}}_{\text{hedging}}, \qquad H(\alpha) = \alpha\,\sigma_e^2 + (1-\alpha)\rho\,\sigma_\pi\sigma_e.$$

Total derivative (allow $\sigma^2_{\Delta r}$ to move with α):

$$\frac{\partial \omega^*}{\partial \alpha} = \underbrace{\frac{(\gamma\kappa - 1)\,H'(\alpha)}{\sigma^2_{\Delta r}}}_{\textbf{(I) direct hedge}} - \underbrace{\frac{E_t[\Delta r_{t+1}] + (\gamma\kappa - 1)H(\alpha)}{(\sigma^2_{\Delta r})^2}\,\frac{\partial \sigma^2_{\Delta r}}{\partial \alpha}}_{\textbf{(II) via risk denominator}}, \qquad H'(\alpha) = \sigma_e^2 - \rho\,\sigma_\pi\sigma_e.$$

Where the "second–order" piece comes from. Let

$$\sigma^2_{\Delta r}(\alpha) = \underbrace{\sigma_e^2}_{\text{FX var}} + \underbrace{a\Big[(i_\$ - i_{\min})^2 + (i^R - i_{\min})^2\Big]}_{\text{rate variance}} + \underbrace{2b\sigma_e\,|i^R - i_\$|}_{\text{FX–rate co-move}}.$$

Then by chain rule

$$\frac{\partial \sigma^2_{\Delta r}}{\partial \alpha} = 2a(i_\$ - i_{\min})\,\frac{\partial i_\$}{\partial \alpha} + 2a(i^R - i_{\min})\,\frac{\partial i^R}{\partial \alpha} + 2b\sigma_e\,\mathrm{sgn}(i^R - i_\$)\left(\frac{\partial i^R}{\partial \alpha} - \frac{\partial i_\$}{\partial \alpha}\right),$$

and

$$\textbf{(II)} = \frac{\omega^*(\alpha)}{\sigma^2_{\Delta r}}\,\frac{\partial \sigma^2_{\Delta r}}{\partial \alpha} = \omega^*(\alpha)\,\frac{\partial \ln \sigma^2_{\Delta r}}{\partial \alpha}.$$

Appendix: $\dfrac{\partial \omega^*}{\partial \alpha}$ and dominance of the risk–denominator term II

Dominance (sufficient) condition for the sign.

$$(\gamma\kappa - 1)\, H'(\alpha) \;>\; \omega^*(\alpha)\, \frac{\partial \ln \sigma^2_{\Delta r}}{\partial \alpha} \quad \Longrightarrow \quad \boxed{\frac{\partial \omega^*}{\partial \alpha} > 0}.$$

Interpretation: the direct hedge channel grows with α (through $H'(\alpha) > 0$), while the denominator effect is the product of (i) the level $\omega^*(\alpha) \in (0, 1)$ and (ii) the semi-elasticity of total risk with respect to α.
Order of magnitude under baseline calibration (illustrative).

$$(\gamma\kappa - 1)H'(\alpha) \;=\; (1.30)\,(\sigma_e^2 - \rho\,\sigma_\pi\sigma_e) = (1.30)\,(0.0324 - 0.0063) \approx 0.0339.$$

For the risk term,

$$\frac{\partial \ln \sigma^2_{\Delta r}}{\partial \alpha} \approx \frac{2a|i - i_{\min}|\,|\partial i/\partial \alpha| + 2b\sigma_e\,|\partial(i^R - i_\$)/\partial \alpha|}{\sigma_e^2 + a[(i_\$ - i_{\min})^2 + \cdots] + 2b\sigma_e|i^R - i_\$|}.$$

With $a = 0.08$, $b = 0.04$, $\sigma_e = 0.18$, $i_\$, i^R \in [0.01, 0.05]$, and moderate $|\partial i/\partial \alpha|$ (quantity-inelastic demand, upward-sloping supply), the numerator is small relative to the denominator dominated by $\sigma_e^2 \approx 0.0324$, so

$$\omega^*(\alpha)\, \frac{\partial \ln \sigma^2_{\Delta r}}{\partial \alpha} \;\ll\; 0.0339.$$

Hence term **(I)** dominates and $\frac{\partial \omega^*}{\partial \alpha} > 0$.

Appendix: $\dfrac{\partial \omega^*}{\partial \alpha}$ and dominance of the risk–denominator term III

Takeaway. Even when $\sigma^2_{\Delta r}$ depends on α through rates, the induced "risk-denominator" channel is formally bounded by $\omega^* \, \partial \ln \sigma^2_{\Delta r} / \partial \alpha$ and is quantitatively dominated by the direct hedge term $(\gamma\kappa - 1)H'(\alpha)$ under standard calibrations. Therefore D^{for} increases with α:

$$\frac{\partial D_t^{\text{for}}}{\partial \alpha} = \frac{A_t^R}{E_t} \frac{\partial \omega^*}{\partial \alpha} + \frac{\partial FX_t^{\$}}{\partial \alpha} > 0.$$

back

Appendix: Derivation of the sovereign-spread function

1. Assume that the ratio of reserves to USD liabilities is a random variable s_t^*. Suppose s_t^* follows a continuous distribution with cumulative distribution function $F_Q(\cdot)$.
2. The sovereign defaults when $s_t^* < 1$. Therefore the default probability is

$$p_t = \Pr(s_t^* < 1) = F_Q(1).$$

3. Conditional on default, investors suffer a loss given default LGD. Under risk-neutral pricing, the spread compensating for expected losses is

$$\phi_t = \mathrm{LGD} \cdot p_t = \mathrm{LGD} \cdot F_Q(1).$$

4. Expanding around a baseline mismatch M_0 yields a local slope

$$\theta_p = \left.\frac{\partial \phi_t}{\partial M}\right|_{M=M_0} = \mathrm{LGD} \cdot f_Q(s_0^*) \cdot \frac{R_0}{M_0^2},$$

where f_Q is the pdf of s_t^* and $s_0^* = R_0 / M_0$.

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Appendix: First-order conditions for reserves and fiscal policy

Central bank.

The central bank minimises

$$\mathcal{L}(FX_t) = E_t \sum_{\tau=t}^{\infty} \beta^{\tau-t} \Big[(i_\tau^{\$} - i_\tau^{R}) FX_\tau + \Pr(FX_\tau < B_\tau^{R,\$}) \, \Lambda \Big].$$

Differentiating with respect to FX_t and imposing the envelope theorem gives

$$(i_t^{\$} - i_t^{R}) = \Lambda \frac{\partial \Pr(\text{crisis})}{\partial FX_t} = \Lambda \, \theta_p \, \frac{1}{B_t^{R,\$}},$$

consistent with the FOC used in the main text.

Fiscal authority.

The government maximises

$$-\frac{1}{2} \sum_{\tau=t}^{\infty} \beta^{\tau-t} \Big[(G_\tau - \bar{g} Y_\tau)^2 + \eta (B_\tau^{\text{gov}} / Y_\tau - \bar{b})^2 \Big]$$

subject to $B_{\tau+1}^{\text{gov}} = (1 + i_\tau^{\$}) B_\tau^{\text{gov}} + G_\tau + TR_\tau - T_\tau$. Taking the derivative with respect to G_t and applying the envelope theorem yields a Euler-type condition linking G_t to $(i_t^{\$} - i_0^{\$})$ and $(B_t^{\text{gov}} / Y_t - \bar{b})$. Linearising around the steady state delivers the feedback rule in the main slides.

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Appendix: Microfoundation of the saving rule I

Goal. Derive S_t^U (net USD saving) from the household Euler + budget constraint.

Setup (real USD, no trend growth). Take the same CRRA utility over the composite C_t as in the main deck:

$$\max_{\{C_t, S_t^U\}} \frac{C_t^{1-\gamma}}{1-\gamma} + \beta \frac{C_{t+1}^{1-\gamma}}{1-\gamma}.$$

Two-period budget (real USD units):

$$C_t + S_t^U = Y_t^U, \qquad C_{t+1} = Y_{t+1}^U + (1 + r_t^{\$}) S_t^U.$$

Define net USD saving (as in the main slide):

$$S_t^U \equiv B_t^{U,\$} + B_t^{R,\$}.$$

Euler. FOC implies

$$u'(C_t) = \beta(1 + r_t^{\$})\, u'(C_{t+1}) \quad \Rightarrow \quad \left(\frac{C_{t+1}}{C_t}\right)^{\gamma} = \beta(1 + r_t^{\$}).$$

Hence

$$C_{t+1} = \kappa_t\, C_t, \qquad \kappa_t \equiv \left[\beta(1 + r_t^{\$})\right]^{1/\gamma}.$$

Appendix: Microfoundation of the saving rule II

Solve for S_t^U. Using $C_t = Y_t^U - S_t^U$ and $C_{t+1} = Y_{t+1}^U + (1 + r_t^\$) S_t^U$:

$$Y_{t+1}^U + (1 + r_t^\$) S_t^U = \kappa_t \left(Y_t^U - S_t^U \right)$$

$$\Rightarrow \quad \boxed{S_t^U = \frac{\kappa_t Y_t^U - Y_{t+1}^U}{(1 + r_t^\$) + \kappa_t}}$$

No-growth specialization. If $Y_{t+1}^U = Y_t^U$:

$$\boxed{\frac{S_t^U}{Y_t^U} = \frac{\kappa_t - 1}{(1 + r_t^\$) + \kappa_t}, \qquad \kappa_t = \left[\beta (1 + r_t^\$) \right]^{1/\gamma}}$$

Borrowing region.

$$S_t^U < 0 \iff \kappa_t < 1 \iff \beta (1 + r_t^\$) < 1 \iff r_t^\$ < \delta_{eff}, \qquad \delta_{eff} \equiv - \ln \beta.$$

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